

Week 9: Project 3 Milestone 1

Data Selection and Project Proposal

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Topic

This project will provide as a proactive tool for both families and healthcare providers in predicting substance abuse in their loved ones. This project will develop a predictive model to help healthcare professionals and family member flag if their patients or loved ones are likely to experience substance abuse of some kind.

Business Problem

The United States has a clear substance abuse problem, and the result of that problem is that it costs government agencies, families, over a trillion dollars every year. In fact, in 2017 alone opioid use disorder alone had an estimated economic cost of over 1.02 trillion dollars (CDC 2021.) In many cases the detection of this substance abuse is very difficult for example in inpatient hospital settings over 70% of cannabis use disorders go undetected (Serowik, K. L., Yonkers, K. A., Gilstad-Hayden, K., Forray, A., Zimbrea, P., & Martino, S. 2021).

This project will attempt to build a predictive model that can help providers and families alike predict if their patients or loved ones are likely suffering from substance abuse based on a data backed model that is easy to predict on given a series of demographic information. Hopefully, this project can help predict those suffering from substance abuse and get them referred to professionals who can help.

This project has serious demand in the marketplace as healthcare providers as well as family have both emotional and financial incentives to properly detect and diagnose substance abuse. Providers would like to detect and diagnose substance abuse at a better rate because it both limits liability and allows them and the hospital network to provide them to professional

care which is lucrative for the healthcare system. Families have an emotion incentive to detect substance abuse disorders because they want the best outcomes for their loved ones.

Datasets

Data for this project will be collected from the Substance Abuse and Mental Health Services Administration, SAMHSA, and their mental health client level data report from 2023. The Dataset includes 40 features and over 7 million rows to train on. Features include demographic information such as age, race, education, career data. Additionally, the dataset includes information on previous mental health disorder diagnosis.

For this project we will use all the Variables that we can with the exception of the variables that identify specific substance abuses, alcohol or opioid binary flag variables, to predict if a person will flap positive for some form of substance abuse. This is done to ensure that we do not have any cases of data leaking in our models.

Methods

To build this model predicting Substance abuse a series of predictive models will be built. The models trained on will be logistic regression, decision tree, random forest, gradient boosted trees as well as an MLP neural network. This diverse group of models trained will ensure that we account for both liner and nonlinear trends as well as the machine learning models ability to capture complex interactions among the data from both demographics and diagnosis.

These models will be evaluated on F2 score. This metric has been chosen due to our emphasis on giving more rate to recall or sensitivity than precision. We want to maximize the

number of true positives detected while minimizing false negatives. All models will be trained on an 80/20 train test split to minimize the effects of overfitting.

Ethical Considerations

There are many ethical considerations of applying this substance abuse predictive model in helping families and healthcare professionals detect substance abuse. The first being that this model may not be 100% accurate in prediction. Meaning that there is always a risk of misidentifying someone either positively or negatively for substance abuse. A false positive could impart unnecessary stress on an otherwise healthy person. Where a false negative leads to patients not receiving the treatment they need to get healthy.

Additionally, this model will train on both race and age demographics. It is likely in this case that the model will treat different demographics differently. As such some demographics could be stuck with a greater portion of unnecessary treatment costs or no have their substance abuse detected given the models predictions.

Challenges/Issues

The first challenge the model will face is that all categorical data in the data set has been coded into numbers. To train on a significant amount of feature engineering will be required to get the data to a place where it can be trained on. It is likely that many dummy variable will need to be created in order to account for these categorical features.

Secondly. Some of the features in the data set are reliant on a previous diagnosis of mental health disorders. If the patient has not been properly diagnosed previously the model may struggle to provide accurate predictions.

References

- Luo, F., Li, M., & Florence, C. (2021). State-level economic costs of opioid use disorder and fatal opioid overdose — United States, 2017. *MMWR. Morbidity and Mortality Weekly Report*, 70(15), 541–546. <https://doi.org/10.15585/mmwr.mm7015a1>
- Substance Abuse and Mental Health Services Administration. (2023). *Mental health client-level data (MH-CLD)*. <https://www.samhsa.gov/data/data-we-collect/mh-cld-mental-health-client-level-data/datafiles/2023>
- Serowik, K. L., Yonkers, K. A., Gilstad-Hayden, K., Forray, A., Zimbrea, P., & Martino, S. (2021). Substance use disorder detection rates among providers of general medical inpatients. *Journal of General Internal Medicine*, 36(3), 668–675. <https://doi.org/10.1007/s11606-020-06319-7>